

B - Personnel Change

分值: 200 分

Problem Statement

The company where Takahashi works has N employees, each assigned an employee number from $1, 2, \dots, N$. There are M departments in the company, called departments $1, 2, \dots, M$.

高桥所在的公司共有 N 名员工, 每位员工都有一个从 $1, 2, \dots, N$ 开始编号的工号。公司共有 M 个部门, 编号为 $1, 2, \dots, M$ 。

Employee i belongs to department A_i this term and will belong to department B_i next term.

员工 i 本学期属于部门 A_i , 下学期将调入部门 B_i 。

For each of departments $1, 2, \dots, M$, find the number of members next term minus the number of members this term.

对于编号为 $1, 2, \dots, M$ 的每个部门, 计算其下学期成员数减去本学期成员数的差值。

Constraints

- $1 \leq N \leq 100$
 - $1 \leq M \leq 100$
 - $1 \leq A_i \leq M$
 - $1 \leq B_i \leq M$
 - All input values are integers.
 - $1 \leq N \leq 100$
 - $1 \leq M \leq 100$
 - $1 \leq A_i \leq M$
 - $1 \leq B_i \leq M$
 - 所有输入值均为整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N M
 A_1 B_1
 A_2 B_2
 $\vdots$
 A_N B_N

Output

Output M lines. The j -th line should contain the answer for department j .

输出 M 行。第 j 行应包含部门 j 对应的答案。

Sample Input 1

```
5 4
1 2
2 1
3 1
2 2
2 4
```

Sample Output 1

```
1
-1
-1
1
```

- For department 1: this term, one employee (employee number 1) belongs; next term, two employees (employee numbers 2, 3) belong.
- For department 2: this term, three employees (employee numbers 2, 4, 5) belong; next term, two employees (employee numbers 1, 4) belong.

- For department 3: this term, one employee (employee number 3) belongs; next term, no one belongs.
 - For department 4: this term, no one belongs; next term, one employee (employee number 5) belongs.
 - 部门 1: 本学期有 1 名员工 (工号 1) ; 下学期有 2 名员工 (工号 2, 3) 。
 - 部门 2: 本学期有 3 名员工 (工号 2, 4, 5) ; 下学期有 2 名员工 (工号 1, 4) 。
 - 部门 3: 本学期有 1 名员工 (工号 3) ; 下学期没有成员。
 - 部门 4: 本学期没有成员; 下学期有 1 名员工 (工号 5) 。
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Sample Input 2

```
10 5
3 2
3 4
1 2
2 2
4 4
3 1
3 4
4 2
3 3
3 2
```

Sample Output 2

```
0
4
-5
1
0
```